

The Fine-Structure Constant as a Size Ratio

H⁻ and the Deuteron as Mirror Proton–Electron Models

Claes Johnson¹

¹KTH Royal Institute of Technology, Stockholm, Sweden , claesjohnson@gmail.com

July 1, 2026

*Mathematical theory and formulation by the author; manuscript developed
in collaboration with Claude (Anthropic). See Section 8.*

Abstract

RealQM [1] and RealNucleus [2] model atoms and nuclei from the *same* proton–electron Coulomb mechanics — charge densities minimising the Coulomb energy, no strong force, no exchange–correlation functional. Their simplest mirror pair is the subject here: the **H⁻ ion** (one proton surrounded by two electrons in two shells) as the *atom* model, and the **deuteron** ²H (one electron caged inside two protons in two shells) as the *nucleus* model — the two related by $p \leftrightarrow e$. Both models take only three inputs: the charge $\pm e$, an electron kinetic coefficient κ , and the proton size R_p — and **no mass m , no speed of light c , no Planck constant $\hbar$** (these never appear separately). The atomic size is $a_0 = 2\kappa/e^2$ (the electrons taper to zero in vacuum, kinetic-set); the nuclear size is R_p (the caged electron is flat, zero kinetic energy, size-set). Their ratio is the fine-structure constant squared,

$$\alpha^2 = \frac{R_p}{a_0}, \quad a_0 = \text{size of the H atom},$$

so $\alpha = \sqrt{R_p/a_0}$ is a pure ratio of two lengths, independent of $m, c, \hbar$. With R_p at the classical-radius scale (≈ 2.8 fm) the relation is exact; at the measured deuteron radius (2.14 fm) it reproduces α to $\sim 25\%$ (Section 4). The two models are independently validated: atomic total energies $< 1\%$ across the periodic table, and the He-4/deuteron binding-energy ratio 13.1 vs. 12.7.

1 Two Coulomb models: RealQM and RealNucleus

The program treats protons and electrons as $\pm$ unit charges (charge densities) interacting by the Coulomb law, with a kinetic penalty on the electron densities; structure is found by minimising the total energy.

- **RealQM** [1] — *atoms and molecules*: electron charge clouds arranged in non-overlapping shells around small proton nuclei, bound by attraction to the nuclear charge. No basis set, no exchange–correlation functional. It reproduces atomic total energies to $< 1\%$ across the periodic table [3].
- **RealNucleus** [2] — *nuclei*: protons bound by shared electrons (neutron = p+e), by the same Coulomb attraction, with *no strong force*. Nested charge-cloud shells give, e.g., the He-4/deuteron binding-energy ratio 13.1 vs. 12.7 experimental.

Both are the same mechanics — charges and a kinetic coefficient — run at two scales. The point of this note is that the two scales are locked together by a single geometric relation, $\alpha^2 = R_p/a_0$.

2 The mirror pair: H^- and the deuteron

Take equal numbers of protons and electrons as $\pm$ charges on the same footing (the starting point of the program; motivated cosmologically in [5]). The simplest three-body bound systems are exact mirrors under $p \leftrightarrow e$:

	atom model (RealQM)	nucleus model (RealNucleus)
system	$\text{H}^- = e + p + e$	deuteron ${}^2\text{H} = p + e + p$
picture	1 proton, 2 electrons in 2 shells	1 electron, 2 protons in 2 shells
centre / surround	proton centred; electrons around	electron caged; protons around
size scale	atomic, $\sim a_0$	nuclear, $\sim R_p$

They are the *same* 3-body Coulomb system — a central charge holding two like-charged partners against their mutual repulsion — differing in nothing but which particle is central. The surrounders set the size: electrons (fluid) $\Rightarrow$ atomic; protons (rigid, of fixed size R_p) $\Rightarrow$ nuclear.

The inputs of both models. Charge $\pm e$; an electron kinetic coefficient κ (taken as a single primitive, not decomposed as $\hbar^2/2m$); and the proton size R_p . **No mass, no c , no $\hbar$** enters separately — only the combination κ , the charge e , and the length R_p . The construction is non-relativistic and electrostatic.

3 The two sizes, and $\alpha^2 = R_p/a_0$

An electron density ψ ($\int \psi^2 = 1$) of size L has energy $E = \kappa \int |\nabla \psi|^2 - Z e^2/L$.

Atomic size (charge-set). Around a proton, an electron must taper to zero in the surrounding vacuum; that varying profile costs kinetic energy $\sim \kappa/L^2$, and minimising $\kappa/L^2 - e^2/L$ gives

$$a_0 = \frac{2\kappa}{e^2} \quad (= \hbar^2/me^2), \quad \text{the size of the H atom.}$$

Nuclear size (size-set). Caged between protons on its own domain with a multiphase *free boundary*, the electron's density does not drop to zero but hands off to the proton density; a flat caged density has $\nabla \psi = 0$, hence *zero kinetic energy*. With no kinetic pressure to spread it, its size is set by the proton's finite size R_p (the cage floor), binding Coulomb at that size.

The ratio. Dividing the two sizes,

$$\boxed{\alpha^2 = \frac{R_p}{a_0} = \frac{R_p e^2}{2\kappa}, \quad \alpha = \sqrt{R_p/a_0}.} \quad (1)$$

Written this way α is a **ratio of two lengths**, obtained by measuring two sizes and dividing: it carries *no m , no c , no $\hbar$* . This is the textbook relation $r_e/a_0 = \alpha^2$ (the classical radius $r_e = e^2/mc^2$ over the Bohr radius) with R_p identified as r_e ; the standard form $\alpha = e^2/\hbar c$ writes the same number through c (and $m, \hbar$), but the ratio itself needs none of them. In standard QM/QED α is the relativistic coupling (amplitude $\sqrt{\alpha}$ per photon vertex; fine-structure splitting $\propto \alpha^2$; $v_1/c = \alpha$); here it is the mass-, c - and $\hbar$ -free proton-to-atom size ratio — the more primitive reading.

4 Computations

Atomic scale. With the electron κ and charge e ,

$$a_0 = \frac{2\kappa}{e^2} = 0.529 \text{ Å} = 52,918 \text{ fm},$$

and RealQM reproduces the H atom at this size with total energy -0.500 Ha (exact).

Nuclear scale and the ratio. Taking R_p at candidate nuclear lengths:

R_p	value (fm)	R_p/a_0	% of α^2	$\sqrt{\cdot} \rightarrow \alpha$
classical radius $r_e = e^2/mc^2$	2.818	5.33×10^{-5}	100%	1/137 (exact)
deuteron rms radius	2.142	4.05×10^{-5}	76%	1/157
proton charge radius	0.841	1.59×10^{-5}	30%	1/251

(True $\alpha^2 = 5.33 \times 10^{-5}$, $\alpha = 1/137.04$.) So with R_p at the classical-radius scale the relation $\alpha^2 = R_p/a_0$ is exact by construction; at the *measured deuteron radius* it reproduces the fine-structure constant to $\sim 25\%$ ($\alpha = 1/157$ vs. $1/137$) — from charge, κ and the proton size alone, with no strong force and no c . (The deuteron is unusually loosely bound and slightly larger than a nucleon, so this is the *scale/ratio* being right, not a precise fit.)

Independent validation of the two models. The mirror models are quantitative on each side:

- *Atom side* — RealQM atomic **total energies** $< 1\%$ across the whole periodic table (all $Z \geq 11$; period-2 p-block B–F the only 3–6% exception) [3].
- *Nucleus side* — RealNucleus **He-4/deuteron binding-energy ratio** 13.1 vs. 12.7 experimental ($\sim 3\%$, nested charge-cloud shells) [2].

So $\alpha^2 = R_p/a_0$ is bracketed by two independent quantitative validations — the atomic scale to $< 1\%$ and the nuclear binding ratio to $\sim 3\%$ — beyond the $\sim 25\%$ size check itself.

5 Mass is a separate quantity

Nothing above used mass: H^- and the deuteron are built from charge and κ . Mass is **inertia** — the localisation (frequency / inverse size) of the matter wave — a *separate, gravitational* quantity, coupling to the electrostatic sector only through energy, $E = mc^2$. Electromagnetism (charge) is an interaction; mass is the inverse size of each matter wave.

6 Complexity from the size asymmetry, and open problems

The proton–electron asymmetry that builds structure is not one of *charge* (charge is exactly balanced, $\pm e$) but of *size*: the proton is fixed-and-small (a rigid anchor), the electron variable. The fixed proton anchors; the variable electron spans scales — shells in the atom, a cage-filling cloud in the deuteron — the nested hierarchy that *is* complexity. And α measures this asymmetry: $\alpha^2 = R_p/a_0$ is the electron’s inverse dynamic range ($1/\alpha^2 \approx 18,800$), so the smallness of α is the engine of complexity.

Open. (i) The nuclear **primitive**: R_p as input length, or equivalently the classical radius $r_e = \alpha^2 a_0$ (electromagnetic mass [4]); α ’s value is reinterpreted, not derived. (ii) Whether the solver realises a flat, zero-kinetic-energy caged electron at the proton scale is not cleanly confirmed. (iii) The equal-footing treatment of p and e vs. the measured mass ratio.

7 Speculative outlook

The same ingredients extend, far more speculatively, to a cosmology — equal-mass $\pm$ charges from two fluctuating potentials, a gravitational bounce with negative-mass dark energy, and a primordial helium fraction $Y \approx 0.245$ — in a companion essay [5]; noted here only for scope.

8 Collaboration

The mathematical theory and the proposal are the author's. The manuscript was developed in dialogue with Claude (Anthropic), which assisted in articulating the argument and its computations. We present the human–AI collaboration transparently as part of the working method.

References

- [1] C. Johnson, *RealQM: Quantum Mechanics as Multiphase 3D Continuum Mechanics*, manuscript and interactive gallery, <https://claes542.github.io/RealMolecule/>.
- [2] C. Johnson, *RealNucleus: nuclear binding from a pure-Coulomb proton–electron model* (neutron = p+e; He-4/deuteron binding ratio from nested charge-cloud shells), manuscript, same repository.
- [3] C. Johnson, *RealQM total energies and ionization energies across the periodic table* (atomic total energies within 1% for $Z \geq 11$), manuscript, same repository.
- [4] C. Johnson, *Many-Minds Relativity* — light as a medium wave, mass as field self-energy ($P = mc \Rightarrow E = mc^2$), and a critique of Einstein's special relativity.
- [5] C. Johnson, *Real Cosmology: a proton–electron universe from two fluctuating potentials* (companion essay), same repository.